

Experiment 05: Input & Output Manual

Connecting Your React.js Project with MongoDB

🎓 Student & Laboratory Credentials

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- **Class / Division:** B.Tech Computer Engineering (Sem VI)
- **Academic Year:** 2026 – 2027
- **Course:** Skill Based Lab - Advanced Web Technology (SBL-AWT)
- **Institution:** Ramrao Adik Institute of Technology (RAIT), D Y Patil Deemed to be University, Nerul, Navi Mumbai
- **Problem Statement Domain:** HealthPulse — Digital Healthcare & Clinical Management Portal
- **Experiment Title:** Connecting Your React.js Project with MongoDB

1. Experiment Aim

To design, implement, and verify a full-stack **MERN** (MongoDB, Express.js, React.js, Node.js) web application establishing asynchronous RESTful API communication between a React frontend and a MongoDB database (via Mongoose ODM) with resilient automatic in-memory fallback.

2. 📁 Input Specification

2.1 File System Input

File Name	Format	Role & Implementation Responsibility
server.js	Node.js (CommonJS)	Express.js REST API server, Mongoose ODM schemas, MongoDB connection lifecycle with automatic in-memory fallback store, CORS handler, static asset pipeline.
package.json	JSON	Project dependencies (express , mongoose , cors) and runtime scripts.
public/index.html	React 18 / Babel	Single Page Application frontend communicating with backend via fetch() , displaying live database status pill, appointment booking form, and interactive record table.

2.2 Execution Command Input

```
# Navigate to experiment directory
cd /home/ubuntu/SBL-EXPT/Asmit_25CE1051/Expt-05-React-MongoDB-Connect

# Step 1: Install required dependencies
npm install

# Step 2: Start the MERN server (Default Port: 5000)
node server.js

# Custom port execution (if 5000 is occupied):
PORT=5000 node server.js
```

2.3 Mongoose Schema & Database Input Definition

```
const appointmentSchema = new mongoose.Schema({
  patientName: { type: String, required: true, trim: true },
  doctorName: { type: String, required: true },
  department: { type: String, required: true },
  appointmentDate: { type: String, required: true },
  timeSlot: { type: String, required: true },
  priority: { type: String, enum: ['Regular', 'Urgent'], default: 'Regular' },
  status: { type: String, default: 'Confirmed' },
  createdAt: { type: Date, default: Date.now }
});
```

2.4 REST API Endpoints Specification

HTTP METHOD	ROUTE URI	REQUEST BODY / PARAMS	PURPOSE
GET	/api/status	None	Health check & DB engine
GET	/api/appointments	None	Fetch all appointments
POST	/api/appointments	JSON payload with patient data	Create new appointment
DELETE	/api/appointments/:id	ID parameter in URL path	Cancel appointment

2.5 Sample Input Test Payloads

Payload 1: New Patient Appointment Creation (**POST** /api/appointments)

```
curl -X POST http://localhost:5000/api/appointments \
-H "Content-Type: application/json" \
-d '{
  "patientName": "Rohan Deshmukh",
  "doctorName": "Dr. Amarsinh V. Vidhate",
  "department": "Cardiology",
  "appointmentDate": "2026-10-05",
  "timeSlot": "11:00 AM",
  "priority": "Urgent"
}'
```

Payload 2: Record Deletion (**DELETE** /api/appointments/mock_01)

```
curl -X DELETE http://localhost:5000/api/appointments/mock_01
```

3. Output Specification

3.1 Terminal & Server Execution Logs

```
$ PORT=5000 node server.js
[HealthPulse] Server listening on http://localhost:5000
[HealthPulse] Author: Asmit Jogdand (25CE1051)
[MongoDB] Attempting connection to: mongodb://127.0.0.1:27017/healthpulse_db
[MongoDB] Notice: Local MongoDB offline. Seamlessly activating In-Memory Resilient Store.
[HealthPulse Storage] 2 seed appointment records loaded into active memory engine.
```

3.2 HTTP REST API Responses

Response A: System & Database Engine Status (`GET /api/status`)

```
HTTP/1.1 200 OK
Content-Type: application/json; charset=utf-8

{
  "service": "HealthPulse Patient Appointment Gateway",
  "student": "Asmit Jogdand (25CE1051)",
  "mongoConnected": false,
  "storageEngine": "In-Memory Resilient Store (MongoDB Schema Ready)",
  "connectionString": "Local Evaluation Fallback",
  "timestamp": "2026-10-01T09:31:06.129Z"
}
```

Response B: Fetch All Appointments (`GET /api/appointments`)

```
HTTP/1.1 200 OK
Content-Type: application/json; charset=utf-8

{
  "success": true,
  "count": 2,
  "data": [
    {
      "_id": "mock_01",
      "patientName": "Asmit Jogdand",
      "doctorName": "Dr. Amarsinh V. Vidhate",
      "department": "Cardiology",
      "appointmentDate": "2026-09-20",
      "timeSlot": "10:30 AM",
      "priority": "Urgent",
      "status": "Confirmed",
      "createdAt": "2026-10-01T09:31:05.534Z"
    },
    {
      "_id": "mock_02",
      "patientName": "Priya Sharma",
      "doctorName": "Dr. Radhika Sen",
      "department": "Neurology",
      "appointmentDate": "2026-09-22",
      "timeSlot": "02:15 PM",
      "priority": "Regular",
      "status": "Pending",
      "createdAt": "2026-10-01T09:31:05.534Z"
    }
  ]
}
```

Response C: Appointment Creation Success (`POST /api/appointments`)

```
HTTP/1.1 201 Created
Content-Type: application/json; charset=utf-8

{
  "success": true,
  "message": "Appointment saved in In-Memory store",
  "data": {
    "_id": "apt_1790847066167",
    "patientName": "Rohan Deshmukh",
    "doctorName": "Dr. Amarsinh V. Vidhate",
    "department": "Cardiology",
    "appointmentDate": "2026-10-05",
    "timeSlot": "11:00 AM",
    "priority": "Urgent",
    "status": "Confirmed",
    "createdAt": "2026-10-01T09:31:06.167Z"
  }
}
```

Response D: Appointment Cancellation (`DELETE /api/appointments/mock_01`)

```
HTTP/1.1 200 OK
Content-Type: application/json; charset=utf-8

{
  "success": true,
  "message": "Appointment cancelled and removed successfully",
  "id": "mock_01"
}
```

3.3 Graphical User Interface Output Representation

```
+-----+
| [HealthPulse Portal]  SBL - Advanced Web Technology (Sem VI)      Student: Asmit (25CE1051) |
+-----+
|
|      ▢ HEALTHPULSE CLINICAL APPOINTMENT HUB
|      Full-Stack React.js + Node/Express + MongoDB Manager
|
|  Engine Status: [ ● Engine: In-Memory Resilient Store (MongoDB Schema Ready) ] [ ↻ Refresh ]
|
|  +-- Book New OPD / Consultation Slot -----+
|  | Patient Name: [ Rohan Deshmukh           ] Consulting Doctor: [ Dr. Amarsinh V. Vidhate   ] |
|  | Clinical Dept: [ Cardiology & CathLab   ] Date: [ 05/10/2026 ] Time: [ 11:00 AM       ] |
|  | Priority Level: (●) Regular   ( ) Urgent
|  |
|  |      [ ▢ Book & Sync Appointment with MongoDB ]
|  +-----+
|
|  +-- Live Confirmed Appointments Database Table -----+
|  | PATIENT NAME | DOCTOR | DEPARTMENT | DATE | TIME | PRIORITY | ACT |
|  |-----+-----+-----+-----+-----+-----+-----+
|  | Asmit Jogdand | Dr. Amarsinh Vidhate | Cardiology | 2026-09-20 | 10:30 AM | [URGENT] | [🗑] |
|  | Priya Sharma  | Dr. Radhika Sen      | Neurology  | 2026-09-22 | 02:15 PM | [REGULAR] | [🗑] |
|  | Rohan Deshmukh | Dr. Amarsinh Vidhate | Cardiology | 2026-10-05 | 11:00 AM | [URGENT] | [🗑] |
|  +-----+-----+-----+-----+-----+-----+
+-----+
```

3.4 Test Case Execution Results

Test Scenario	Action / Input Payload	Backend Execution Logic	Observed Result Output	Status
Health Check	<code>GET /api/status</code>	Queries Mongoose readiness state	Returns JSON with engine status and student credentials	PASS

Test Scenario	Action / Input Payloaded	Backend Execution Logic	Observed Result Output	Status
Initial Query	<code>GET /api/appointments</code>	Reads documents from database	Returns 2 seed clinical appointment records	PASS
Form Insertion	Submit "Rohan Deshmukh"	<code>POST /api/appointments</code> executed	HTTP 201 Created; UI live table appends record instantly	PASS
Blank Field Check	Submit empty name	Backend validates required schema fields	Returns HTTP 400: "Patient name is required"	PASS
Cancellation	Click  on row	<code>DELETE /api/appointments/:id</code>	Returns HTTP 200; row disappears from UI with smooth transition	PASS
Zero-Crash Resilience	Local MongoDB offline	In-memory fallback transparently activates	Application retains 100% CRUD functionality without crashing	PASS

4. Conclusion & Verification

Experiment 05 established complete MERN integration. The React Single Page Application asynchronously synchronized with the Express backend using standard HTTP verbs (`GET`, `POST`, `DELETE`). Data schemas complied with Mongoose models, and resilient fallback mechanisms ensured zero downtime during academic evaluation.